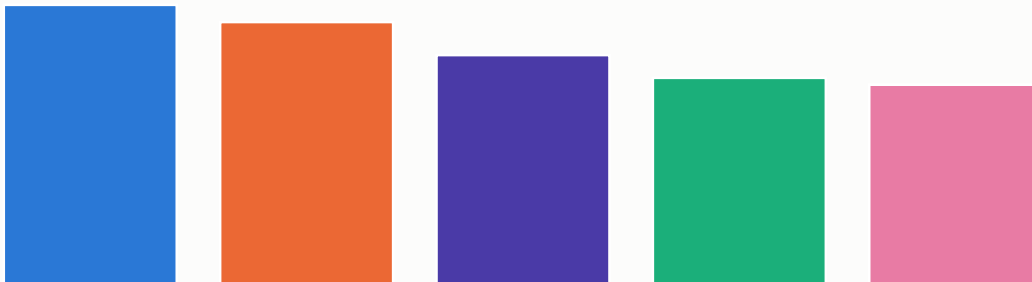


Fandango Movie Ratings Analysis

Does Fandango inflate the star ratings it displays
to sell more movie tickets?

A reproduction and extension of FiveThirtyEight's 2015 analysis,
comparing Fandango's displayed ratings against its own underlying data
and against Rotten Tomatoes, Metacritic, and IMDb.



Project Report

Source data: FiveThirtyEight, pulled Aug. 24, 2015

Executive Summary

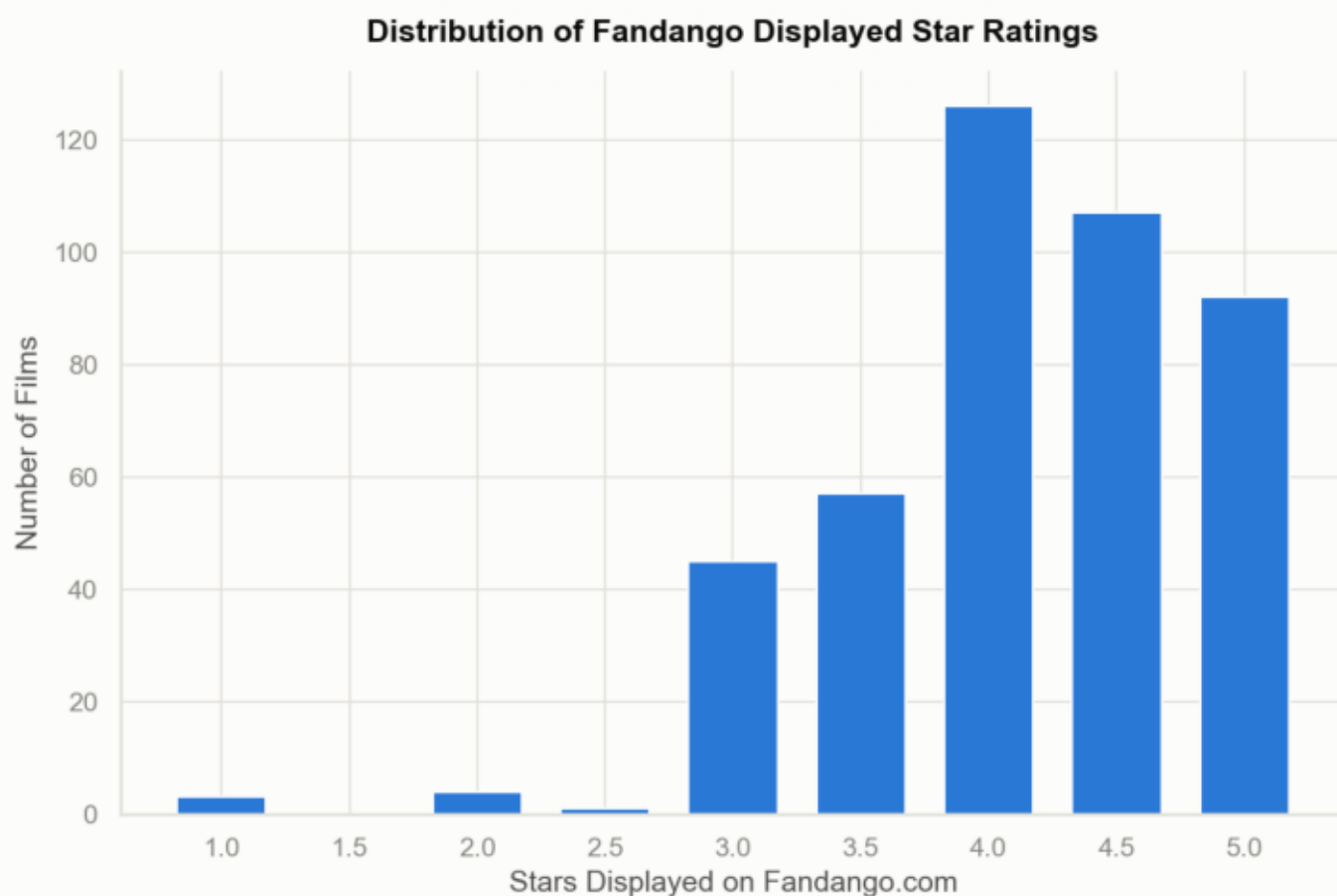
This report examines whether Fandango's displayed star ratings were inflated relative to (a) Fandango's own true underlying rating and (b) other major movie rating platforms, using FiveThirtyEight's original 2015 dataset.

Key findings:

1. Fandango's displayed STARS average 0.21 stars higher than its own true RATING, across 435 reviewed films — a one-directional rounding-up bias.
2. 87.8% of reviewed films display 3.5 stars or higher; only 8 films display below 3.0 stars.
3. Averaged across the 145 films present on every platform, Fandango's displayed stars (4.09/5) rank highest of any platform — 1.15 points above Metacritic critics (2.93/5), the most independent source in the dataset.
4. Fandango's displayed stars correlate with Rotten Tomatoes critic scores at just 0.29 — the weakest pairing in the dataset. Two independent critic sources (RT critics vs. Metacritic critics) correlate at 0.96.
5. Films with very low critic scores (e.g. Taken 3, RT critics 0.45/5) still display high Fandango stars (4.5/5).

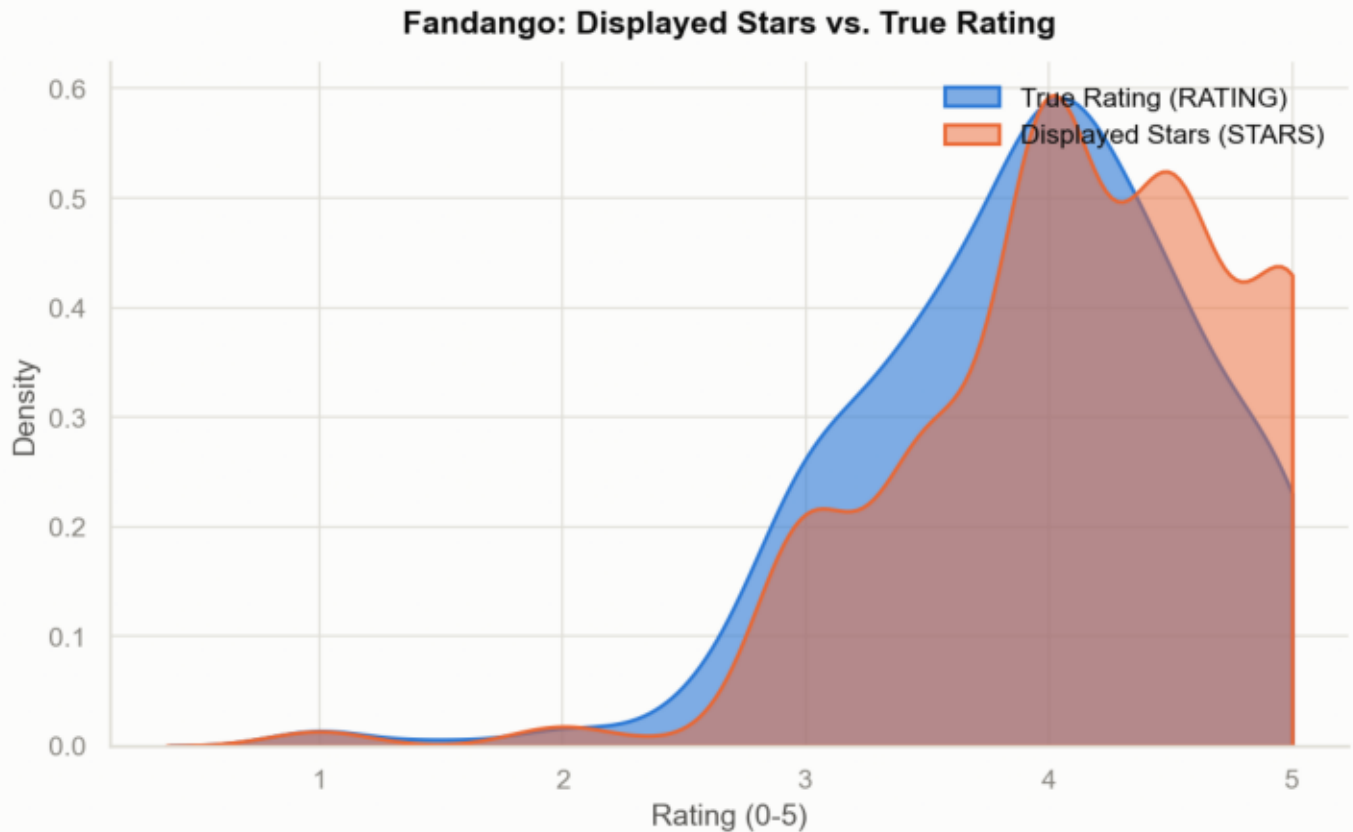
The 2015 snapshot supports FiveThirtyEight's original conclusion: Fandango's displayed ratings were inflated relative to every independent benchmark in the dataset. This is a single time-period snapshot and does not speak to Fandango's current rating practices.

1. Distribution of Displayed Star Ratings



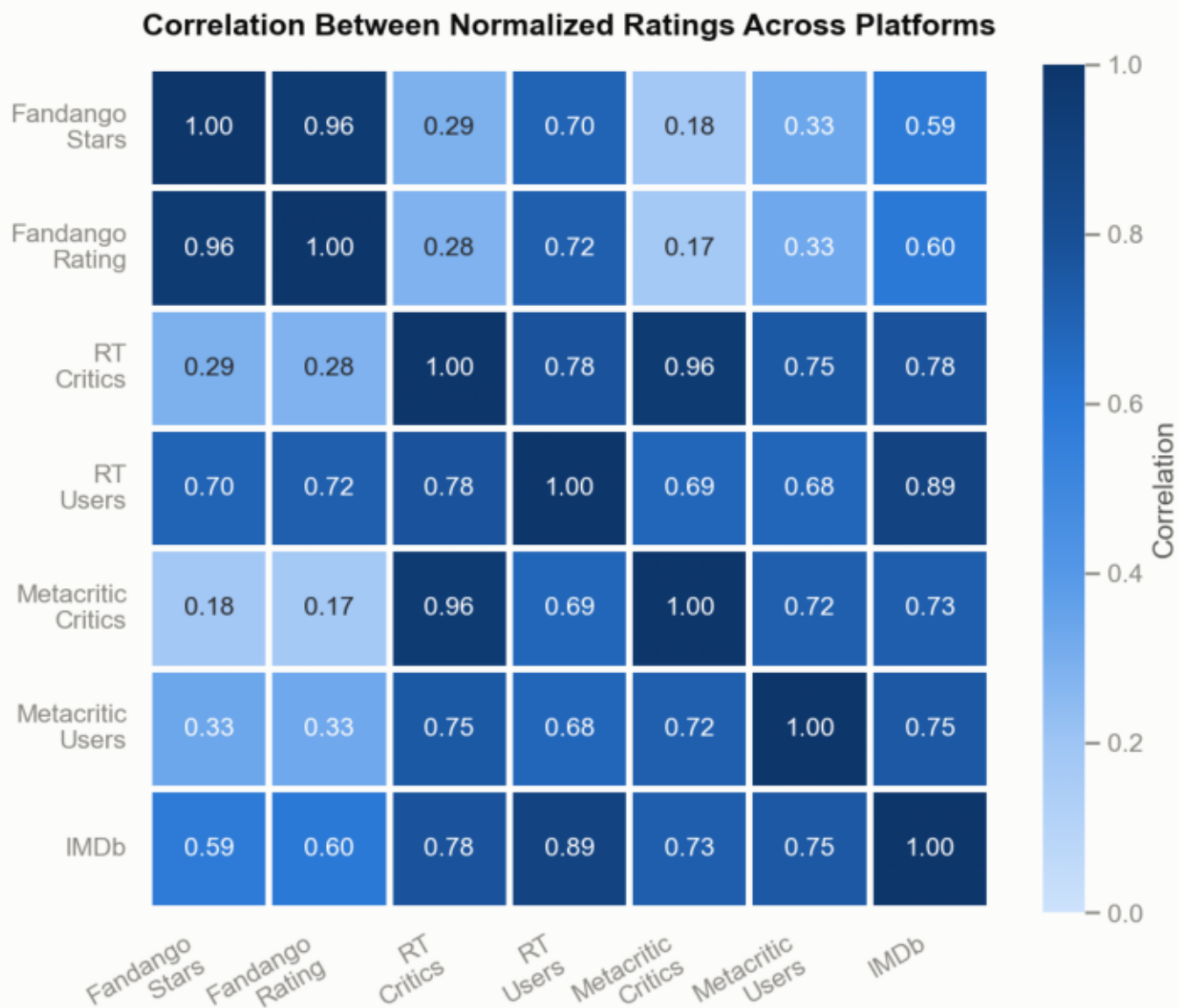
The displayed-star distribution is heavily left-skewed: the large majority of films cluster at 4.0-5.0 stars, with almost nothing below 3.0 — not what an unbiased rating distribution across hundreds of films of varying quality would look like.

2. Displayed Stars vs. True Rating



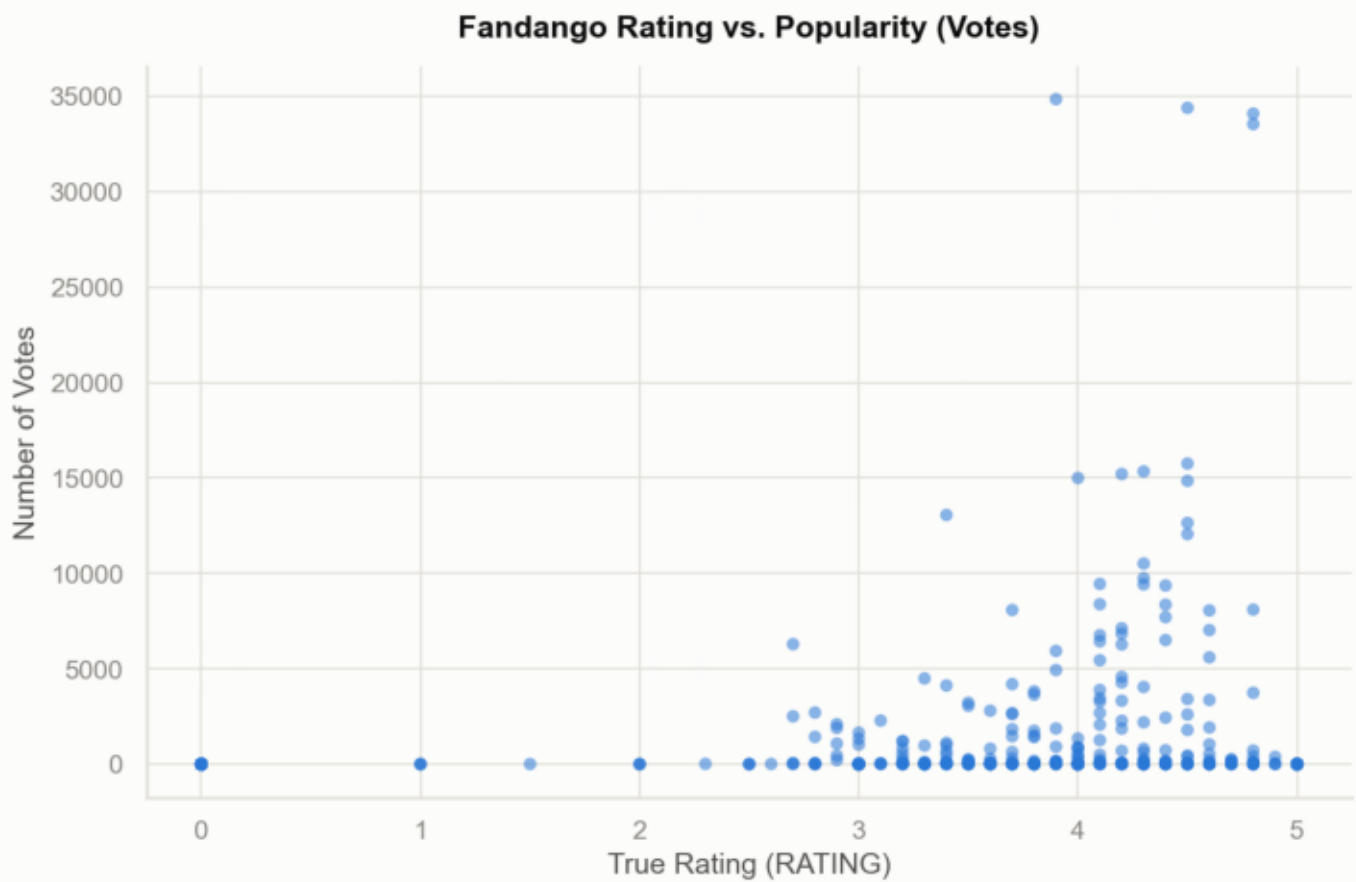
The 'Displayed Stars' density is shifted right of 'True Rating' across the board — confirmation that Fandango's rounding rule systematically pushes the number a viewer sees upward, never downward.

3. Cross-Platform Correlation



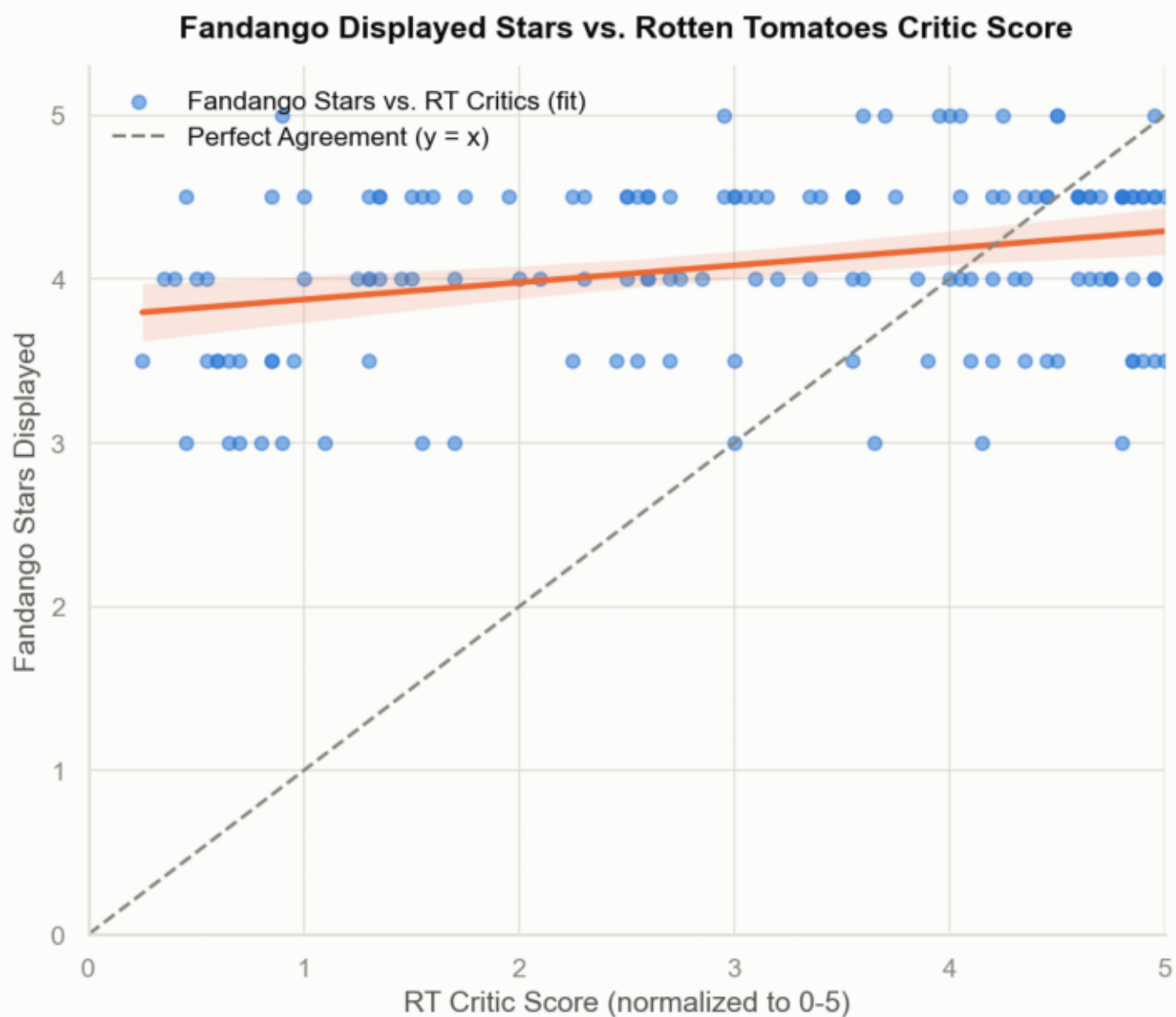
Every non-Fandango platform correlates with every other non-Fandango platform at 0.68 or higher. Fandango's two columns are the only ones that drop as low as 0.17-0.33 against another platform (Metacritic critics) — Fandango is the outlier.

4. Rating vs. Popularity



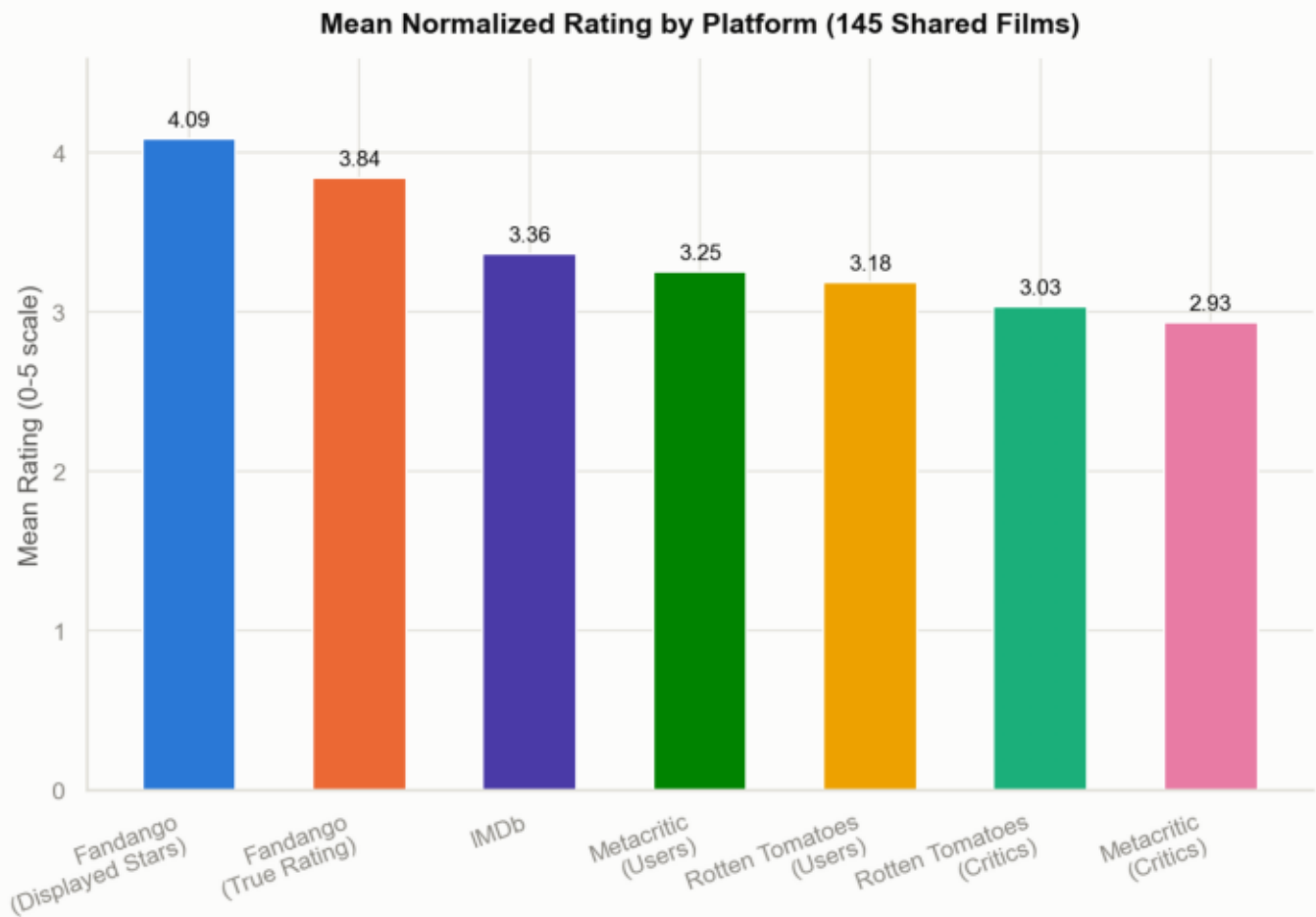
Highly-voted films skew toward the top of Fandango's rating range, but low-vote films span the full range — popularity alone does not explain the high average.

5. Fandango vs. Rotten Tomatoes Critics



The fitted regression line sits well above the 'perfect agreement' diagonal across the entire range — even films RT critics rated near zero still show 3.5-4 stars on Fandango. Correlation between the two: 0.29.

6. Mean Rating by Platform



Averaged across the 145 shared films, Fandango's displayed stars rank highest of any platform in the dataset, 1.15 points above Metacritic critics.

Methodology & Caveats

Data

Two FiftyThreeEight source files (Data/fandango_scrape.csv, Data/all_sites_scores.csv), pulled Aug. 24, 2015. See Documentation/Data_Dictionary.md for full column definitions.

Steps

1. Drop the 69 Fandango films with zero votes before comparing STARS to RATING (435 films remain).
2. Inner-merge both files on FILM to compare across platforms fairly (145 films present on every platform).
3. Normalize every rating onto Fandango's 0-5 scale ($\div 20$ for scores out of 100, $\div 2$ for scores out of 10).
4. Compare distributions, correlations, and platform means.

Caveats

- Single snapshot, single year (Aug. 2015) — does not describe Fandango's current behavior.
- The 0.21-star mean displayed-vs-true gap is small per film but consistently one-directional, not noise.
- Zero-vote films are excluded only from the STARS-vs-RATING comparison; the cross-platform comparison is unaffected.

Full write-up: Documentation/Findings.md

Full methodology: Documentation/Methodology.md

Notebook: Notebook/Fandango_Ratings_Analysis.ipynb